Aurora X2 Smart Water Meter: Technical Specification

Product datasheet · Model AX2-15 and AX2-20 · Revision C · August 2026

> Fictional document created for a live RAG demo. Names, numbers and events are invented.

1. Overview

The Aurora X2 is an ultrasonic smart water meter for homes and small buildings. It has no moving parts, measures flow in both directions, and reports readings over LoRaWAN or NB-IoT to the utility's head-end system.

2. Specifications

Parameter	AX2-15	AX2-20
Pipe size	15 mm (DN15)	20 mm (DN20)
Nominal flow (Q3)	2.5 m³/h	4.0 m³/h
Minimum flow (Q1)	10 litres/hour	16 litres/hour
Accuracy	±1.5% in the upper flow zone, ±3% in the lower zone	
Battery life	12 years at one reading per hour (lithium, not replaceable)	
Protection	IP68, submersible to 1.5 m for 30 days	
Operating temperature	+1 °C to +50 °C (water)	
Radio	LoRaWAN 865–867 MHz (India band) or NB-IoT	
Data storage	Hourly readings for 18 months on the meter	

3. Alarms and error codes

Code	Meaning	What to do

E01	Leak: continuous flow for more than 24 hours	Check taps, toilets and underground pipes; alarm clears after 1 hour of zero flow
E02	Reverse flow above 20 litres	Inspect for a missing non-return valve
E03	Empty pipe or air in the pipe	Normal after supply interruptions; clears when water returns
E04	Burst: flow above 1.5 × Q3 for 30 minutes	Close the inlet valve and call the utility
E05	Tamper: magnetic field or cover opened	Utility inspection within 72 hours
E06	Battery below 10%	Plan meter replacement within 6 months
E07	Water temperature out of range	Check for hot water back-feed
E08	Radio has not joined the network for 7 days	Check antenna position and coverage
E09	Firmware signature check failed	Update rejected; the meter keeps running the previous firmware

4. Installation

1. Flush the pipe before installing to remove debris.
2. Install with the arrow pointing in the flow direction. Horizontal or vertical mounting is allowed.
3. Keep at least 5 × pipe diameter of straight pipe before the meter and 3 × after it.
4. Do not install at the highest point of the pipe, where air collects (causes E03).
5. Activate by touching the optical port with the service wand for 3 seconds; the display shows "JOIN" until the radio connects.

5. Security

Readings are encrypted with AES-128 using keys unique to each meter. Firmware updates over the air are signed; the meter rejects unsigned or tampered images and reports E09. The optical service port requires an installer PIN, which is locked for 15 minutes after 3 wrong attempts.

6. Warranty

Aurora provides a 5-year warranty on the meter and a 12-year pro-rata warranty on the battery.